

Staged modelling progression

simple MDP → richer actions / state / risk

[Iteration 1: Simple Baseline]

- Discrete Actions {Hold, Buy, Sell}
- Binary Holdings $h_t \in \{0, 1\}$
- Simple Step-Return Reward



[Iteration 2: Multi-Sizing & Shorting]

- Expanded Discrete Actions $\{-2, -1, 0, +1, +2\}$
- Short Selling Support ($h_t \in \{-1, 0, 1\}$)
- Volatility-Penalized Reward



[Iteration 3: Continuous Action Sizing]

- Continuous Action Space $a_t \in [-1.0, +1.0]$
- Dynamic Portfolio Capital Allocation
- Feature Vector State (Technical Indicators)



[Iteration 4: Multi-Asset Portfolio Allocation]

- Vector Actions $w_t = [w_1, w_2, \dots, w_N]^T$
- Asset Covariance Matrix in State Space
- Drawdown Penalties & Risk Constraints

Iteration 1 = Friday baseline · later iterations earned one at a time